

A. POWER BI — DATA CLEANING QUESTIONS

Use only Power Query + Data Modeling concepts.

Transaction ID

1. Identify and remove duplicate Transaction ID values using Power Query.
 2. Check whether all Transaction IDs follow the same prefix ("TXN_") using conditional column.
 3. Create a rule to flag invalid or blank Transaction IDs.
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Item

4. Identify inconsistent item names (uppercase/lowercase, spelling variations).
 5. Create a Cleaned Item column using Format > Capitalize Each Word.
 6. Replace invalid item values (e.g., UNKNOWN, blank).
 7. Detect items that appear unusually few times (potential misspellings).
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Quantity

8. Identify and replace negative or zero quantities.
 9. Convert Quantity to Whole Number type.
 10. Create a custom column to flag suspicious quantity values.
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Price Per Unit

11. Convert Price Per Unit to Decimal Number type, fixing errors.
 12. Identify rows where Price contains "ERROR" or non-numeric text.
 13. Create a conditional column marking invalid prices (≤ 0 or extremely high).
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Total Spent

16. Validate that Total Spent = Quantity * Price Per Unit.
17. Create a custom column to recalculate Total Spent.

18. Identify rows where original Total Spent does not match calculated value.
 19. Replace invalid values (e.g., "ERROR") using Replace Errors in Power Query.
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Payment Method

20. Standardize Payment Method values using Transform > Format > Capitalize.
 21. Replace "UNKNOWN", blank, or rare methods with "Other".
 22. Create a lookup table for Payment Methods and map standardized values.
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Location

23. Identify invalid Location entries such as "UNKNOWN".
 24. Standardize locations (In-store, Takeaway, Delivery).
 25. Create a conditional column to flag suspicious locations.
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Transaction Date

26. Convert Transaction Date to Date type.
 27. Identify and filter out invalid or future dates.
 28. Extract Year, Month, Day, Weekday using Date functions.
 29. Create a "Business Month" column (Month Name + Year).
 30. Detect missing dates and decide how to handle them.
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Dataset-Level Cleaning

31. Remove rows with multiple critical issues (using Keep Rows > Remove Errors).
32. Create a Data Quality Flag column capturing all issues (null, invalid, mismatch).
33. Ensure all numeric fields have correct data types (Whole, Decimal).
34. Identify inconsistent price patterns for each item using Group By.
35. Standardize column names to follow naming conventions (no spaces).

B. POWER BI — DATA VISUALIZATION QUESTIONS

All visuals must be created only with the actual dataset fields.

Item-Level Visuals

- 36. Create a bar chart showing total revenue (Total Spent) by Item.
- 37. Create a column chart showing total Quantity sold per Item.
- 38. Add a table visual showing Item-wise average Price Per Unit.
- 39. Create a stacked bar showing Quantity by Item and Location.
- 40. Create a heatmap showing Item on rows and Payment Method on columns (value = total spent).

Payment Method Analysis

- 41. Create a pie chart showing share of Payment Methods used.
- 42. Create a bar chart comparing average Total Spent per Payment Method.
- 43. Show quantity distribution by Payment Method using a clustered column chart.
- 44. Build a donut chart showing number of transactions by Payment Method.

Location Insights

- 45. Create a bar chart showing total revenue by Location.
- 46. Create a table visual listing average price per location.
- 47. Create a clustered column chart of Quantity by Location.
- 48. Create a 100% stacked bar chart comparing Payment Method distribution by Location.

Date-Based Visuals

- 49. Create a line chart showing daily Total Spent over time.
- 50. Create a bar chart showing total transactions per month.

- 51. Create a line chart showing trend of Quantity over time.
 - 52. Create a KPI visual showing Month-To-Date Total Sales.
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Advanced Visual Insights

- 53. Build a scatter plot of Price Per Unit vs Total Spent.
 - 54. Build a scatter plot of Quantity vs Total Spent.
 - 55. Create a decomposition tree with Total Spent broken down by Item > Location > Payment Method.
 - 56. Add a funnel chart showing top-selling items ranked by revenue.
 - 57. Create a matrix showing Item (rows) × Location (columns) with total quantity.
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Business Insights

- 58. Create a Top 5 Items by Revenue bar chart using Top N filter.
- 59. Create a Bottom 5 Items by Quantity sold visual.
- 60. Create a card visual showing Total Number of Transactions.
- 61. Create a card showing Total Revenue.